

Original article

Energy, Economic, Environment (3E) analysis of WT-PV-Battery autonomous hybrid power plants in climatically varying regions

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ABSTRACT

Electricity production via renewable energies is considered to be a suitable solution in the short and long term, since fossil fuels are doomed to exhaustion. The aim of this study is to conduct energy, economic and environmental (3E) analysis for an autonomous hybrid renewable energy system (HRES) composed of a wind and solar photovoltaic systems. In this article, HOMER Pro software is adopted for economic evaluation and optimization. The simulations were carried out for a total of 24 selected cities distributed in different climatic regions for a comprehensive assessment approach. The meteorological data for these cities was generated by the NASA platform. The daily energy load of 319.35 kWh/day was used for a typical village in the Fez-Meknes region with a peak power of 30.89 kW. The examined levelized cost of energy ranged between 0.32 \$/kWh and 0.171 \$/kWh. The lowest levelized cost of energy was predicted for Eddakhla city. The energy shares in this site were to be 25.6% and 74.4%, respectively for the wind turbine system and the PV field. The HRES saves 128.21 tonnes of CO₂ release in the city of Eddakhla. This type of projects will enable Morocco to increase its energy independence and supports its national strategy. Furthermore, these projects can promote the socioeconomic development in remote and isolated sites and can be widespread in the African continent, especially with the decreasing costs of renewable energy equipment.

Introduction

Presently, the deployment of energy from renewables is becoming a very fundamental aspect in Morocco's strategy (see Fig. 1). Wind and solar energy are among the main resources of renewable energy [6,26], and have been developed all over the world during the last decade. The production of electricity from renewable energy resources has numerous advantages compared to conventional resources such as free availability, inexhaustibility and environment preservation [19,29].

More specifically, hybrid renewable energy system (HRES) has several advantages such as flexibility, reliability, low pollution and are very adapted to deliver electrical power to remote communities, cities and towns [28].

Several research works investigated the performances of various combinations of autonomous HRES. Some of the most recent works are

reviewed below.

An optimization for integrated stand-alone hybrid solar PV tracking and diesel generator power system in Khorfakkan city, United Arab Emirates was sized based on techno-economic and environmental aspects [41]. Both lowest cost of energy (COE) and high renewable fraction objectives were investigated.

An analysis on the basis of environmental and economic factors has been brought to optimal design hybrid renewable energy system (PV/Wind) and has shown that 24% of CO₂ emissions can be reduced compared to the diesel system alone [29].

Ghenai et al. [42] performed technico-economic analysis to meet the desired electric load of a residential community located in a desert region. An off grid hybrid solar PV/Fuel Cell power system was used in their study based on integrated modeling, simulation, and optimization approach along with development of dispatch control strategies.

Suresh et al. [39] carried out an electrification study in India on three

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Nomenclature

P_{gen}	power produced by the sub-system [kW]
P_{Dem}	power demand [kW]
P_{WT}	Power produced by Wind turbine [kW]
P_{PV}	Power produced by PV panels [kW]
P_{batt}	Power stored in batteries [kW]
N_{PV}	Number of PV panels [-]
N_{PV_s}	Number of PV panels connected in series [-]
N_{PV_p}	Number of PV panels connected in parallel [-]
f_{rated}	Derating of PV panels [-]
K_T	temperature coefficient [-]
T_C	temperature in the surface of the panels [°C]
T_{ref}	temperature of the solar cells under standard conditions [25 °C]
E_{PV}	Energy produced by PV panels [kWh]
A	represent the strip swept by the turbine blades [m ²]
C_{pc}	mechanical efficiency of the wind turbine [-]
v	wind speed [m/s]
ρ	air density [1.225 kg/m ³]
E_{WT}	Energy produced by wind turbine [kWh]
A_{ts}	area swept by the wind turbine [m ²]

D_{ele}	daily electrical charge [kWh/d]
N_a	number of days of autonomy [d]
η_B	Efficiency of battery [%]
η_{inv}	Efficiency of inverter [%]
D_d	depth of the battery discharge [-]
E_{gen}	Energy produced by HRES [kWh]
E_{excess}	excess of energy production [kWh]
E_D	Energy demand [kWh]
COE	Cost Of Energy [\$/kWh]
CRF	Capital Recovery Factor [-]
NPC	Net Present Cost [€]

Abbreviations

HRES	Hybrid Renewable Energy System
DC	Direct Current
LI	Lithium-ion
NASA	National Aeronautics and Space Administration.
NREL	National Renewable Energy Laboratory
O&M	Operating and Maintenance
PV	Photovoltaic
WT	Wind Turbine
Yr	year

villages in the Kollegal block of the Chamarajanagar district using two methods: Homer pro simulation and the genetic algorithm. They found that the optimal HRES solution consisted of biogas/biomass/solar/wind/fuel cell and battery with COE equal to 0.163 \$/kWh. Singh et al. [38] have introduced an optimal sizing of PV/wind/battery using Homer software for Kavaratti Island, the metropolis of Lakshadweep. They demonstrated that the proposed system allows to deduce advantages such as low energy cost, elimination of fictitious load compared to the existing system and lower emissions, the cost of electricity and the

net present cost (NPC) was found to be 0.10995 \$/kWh and 15,039,705 \$, respectively, over a lifetime of 25 years. Bhakta et al. [10] studied the technical and economic performance of autonomous HRES for the Lakshadweep Islands of India using HOMER software. The authors found that HRES composed of wind-PV-battery was more financially competitive compared to the configuration composed of PV-battery or wind-battery. Razmjoo et al. [35] analyzed and evaluated technoeconomically the viability of hybrid energy system (HES) composed of Wind-Diesel-PV for two cities in Iran Zabol and Zahak using Homer

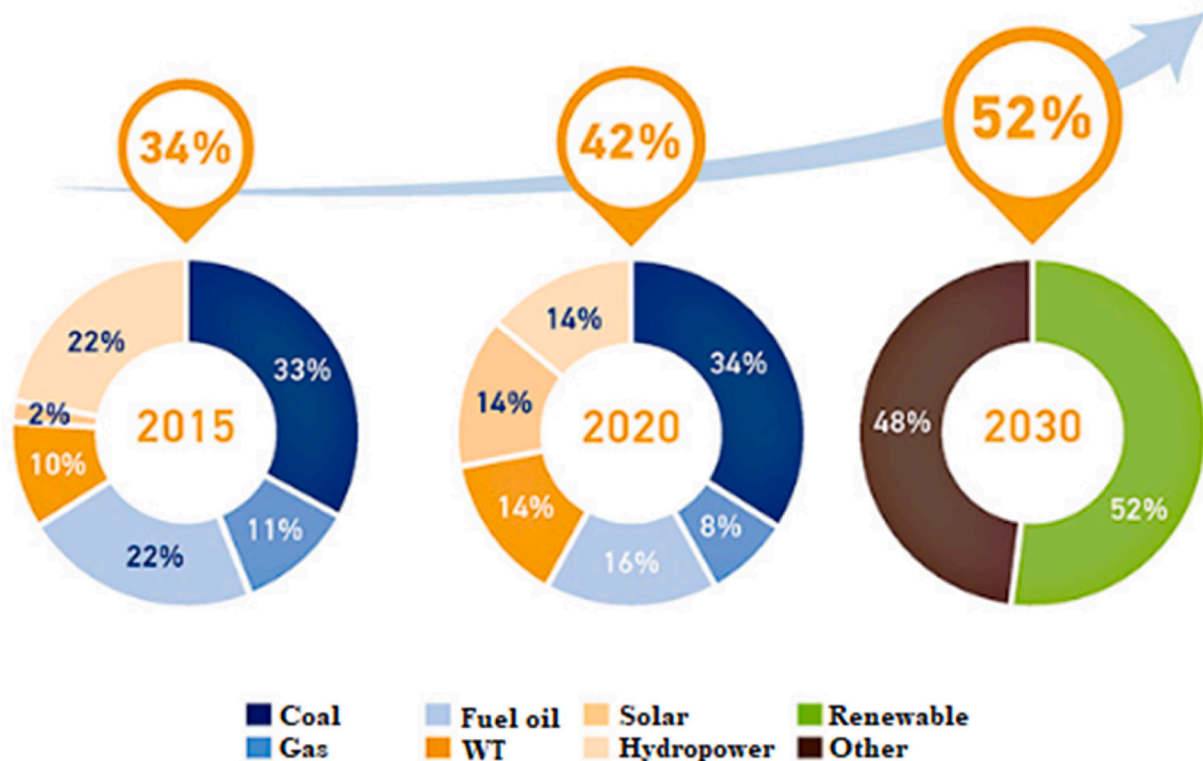


Fig. 1. Increase in the Moroccan energy mix between 2008 and 2030.

software. They found that the capacity of electrical power production by the photovoltaic field for Zabol and Zahak were 1700 kWh/yr and 1669 kWh/yr, respectively, while the electrical power production by wind turbine for Zabol and Zahak were 9036 kWh/yr and 7263 kWh/yr, respectively. Saheli et al. [7] used Homer software to optimize and study the viability of the wind-PV-diesel system for a house in Winnipeg in Canada whose daily and peak load was 16 kWh and 1.9 kW, respectively. The authors compared several systems using an optimization approach. The optimal COE and NPC for this study were 0.444 \$/kWh and \$ 33,110, respectively. Allouhi et al. [3] examined an optimal configuration of PV solar pumping system to meet the water needed in five houses in Moutawakila, Birtamtam in Morocco. The results have proven that the PV system integrating maximum power point tracking (MPPT) is more financially viable than direct coupling in terms of water pumping. Razmjoo and Davarpanah [36] conducted an economic and technical analysis of a hybrid system composed of WT-PV-Diesel Generator in Semirom in Iran. The results elucidated that the proposed design is capable of producing a significant amount of required energy. The optimal sizing corresponded to a 15 kW photovoltaic generator, 13,603 Surrette 6CS25P battery, 1 kW generator, 2 WES 5 Tulipo wind turbines, and 1 kW of Inverter and rectifier.

Kasaeian et al. [23] have brought a technico-economic study for Bandar Dayyer of two hybrid systems. The total energy production of by hybrid systems is found in the results is 470,176 kW of which 22,409 kW are supplied by PV - generic system and 447,767 kW are supplied by PV - diesel - generic system.

Baseer et al. [9] focused on the techno-economic analysis of hybrid wind-PV-battery and wind-PV-diesel-battery for three zones in Saudi Arabia. They concluded that the hybrid photovoltaic/wind system is more appropriate compared to the system which only includes wind.

Several works have been presented in the literature to study the feasibility of developing a HRES to supply electrical energy to various end-uses applications. However, the feasibility study on the development of a HRES has not yet been performed for a wide range of cities with different climatic conditions, especially in Morocco. The development of this type of system is crucial to support the massive development of HRES in Morocco. To sum up, the objectives of this study as can be listed as:

- Performing a comprehensive study for autonomous HRESs for a total of 24 cities (see Table 1). These cities are distributed in twelve regions based on the geographical classification of Morocco.
- Evaluating the renewable energy potential resources for Morocco country based on the six climatic zones announced by Moroccan authorities recently that present the twelve regions and 24 cities.
- Providing the Moroccan policymakers with all supporting information in order to take decision for installing optimal wind-photovoltaic electrification systems in rural areas based on multi-criteria objectives.

This manuscript is structured as follows. Section “Methodology” describes the followed methodology. The profile of the electricity demand, architecture of the hybrid system components, simulation process and key financial and environmental indexes are shown. Section “Results and discussion” presents the results and the discussion starting with the optimal configuration found, the electrical production of the two energy sources in the 24 cities studied, mitigation potential of CO₂ emissions, cost reduction effect and social dimension before giving the main conclusions of the study in Section “Conclusion”.

Methodology

The methodology followed in this article is depicted in Fig. 2. The description is mentioned in the following sub-sections.

Table 1

A global overview of the cities concerned by HRES.

Region	City	Climatic zone	Latitude	Longitude	Altitude
Tangier-Tetouan-	Chefchaouen	2	35°10'1" N	5°16'1" W	564 m
Al Hoceima	Al Hoceima	3	35° 15' 0" N	3° 55' 9" W	74 m
Oriental	Nador	2	35°10'1" N	2°55'7" W	19 m
	Oujda	3	34°40'9" N	1°54'0" W	549 m
Fez-Meknes	Sefrou	3	33°49'8" N	4°50'1" W	809 m
	Ifrane	4	33°31'4" N	5°60'7" W	2019 m
Rabat-Sale-Kenitra	Sidi Slimane	3	34°15'7" N	5°55'4" W	37 m
	Rabat	1	33°58'3" N	6°51'0" W	1113 m
Beni Mellal-Khenifra	Azilal	3	31°57'9" N	6°34'4" W	1351 m
	Khenifra	4	32°56'0" N	5°39'7" W	13 m
Casablanca-Settat	El Jadida	1	33°13'9" N	8°30'0" W	10 m
	Settat	3	32°59'5" N	7°37'3" W	365 m
Marrakech-Safi	Essaouira	1	31°30'5" N	9°45'6" W	129 m
	Al Haouz	5	31°17'7" N	7°52'3" W	1012 m
Darâa-tafilalet	Errachidia	6	31°55'6" N	4°25'7" W	1039 m
	Tinghir	6	31°31'2" N	5°31'8" W	1300 m
Souss-Massa	Agadir	1	30°25'7" N	9°35'9" W	31 m
	Tata	6	29°45'1" N	7°58'5" W	670 m
Guelmim-Oued Noun	Assa-zag	6	28°80'4" N	9°43'4" W	442 m
	Tantan	1	28°26'3" N	11°50'9" W	51 m
Laayoun-Sakia El Hamra	Laayoun	1	27°90'0" N	13°11'9" W	68 m
	Essemara	6	26°44'5" N	11°40'7" W	188 m
Eddakhla-Oued Eddahab	Eddakhla	1	23°43'3" N	15°56'1" W	5 m
	Oued Eddahab	6	22°44'0" N	14°17'2" W	400 m

Electrical load

In this work, the electrical consumption was evaluated based on the survey carried out on the energy bills analysis for the whole year of 2019 in a Moutawakila village, Birtamtam, in the Fes-Meknes region. Fig. 3 illustrates the monthly electric requirements for the village. According to this figure, the consumption of electrical energy is the highest in July (9900 kWh), while it is the lowest in January month (5400 kWh). In Moutawakila village, the number of people in households varies between 4 and 8 peoples. Fig. 4 shows the average daily electrical energy consumption profile based on the Moroccan National Electricity Office (MNEO). It should be stressed here that the seasonal and annual variations of electrical energy consumption were considered negligible due to the absence of industrial and intense commercial activities in this village. This assumption was checked by gathering multi-year data of electric loads recorded by the MNEO in the studied region. The energy use of the village was found to show marginal fluctuation over the years.

From Fig. 4, it is evident that the consumption of electricity is low between 2:00 AM and 6:00 AM. After that, the consumption of electricity increases from 7:00 AM to 8:00 AM then it decreases at 9:00 AM.

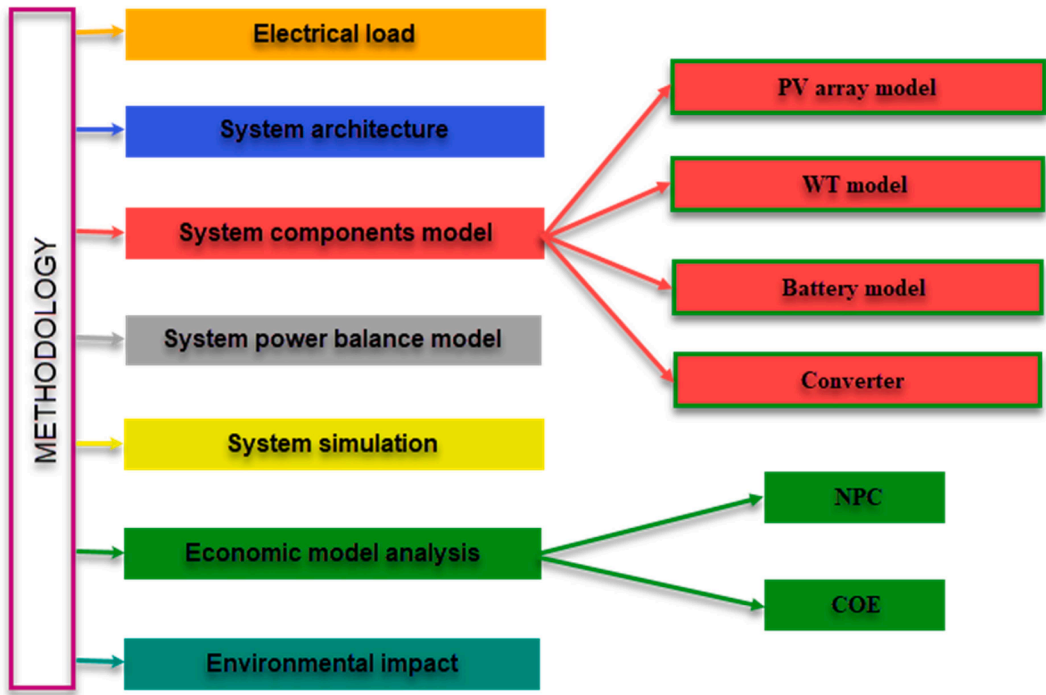


Fig. 2. Methodology adopted in this work.

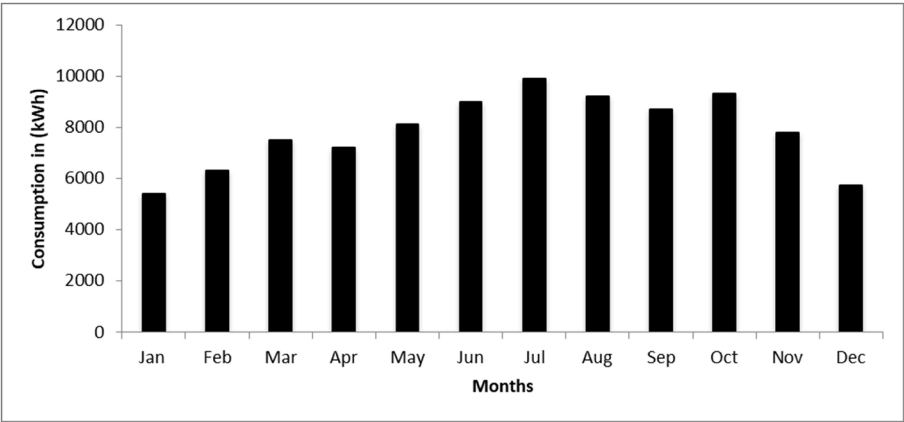


Fig. 3. Monthly average load of the village.

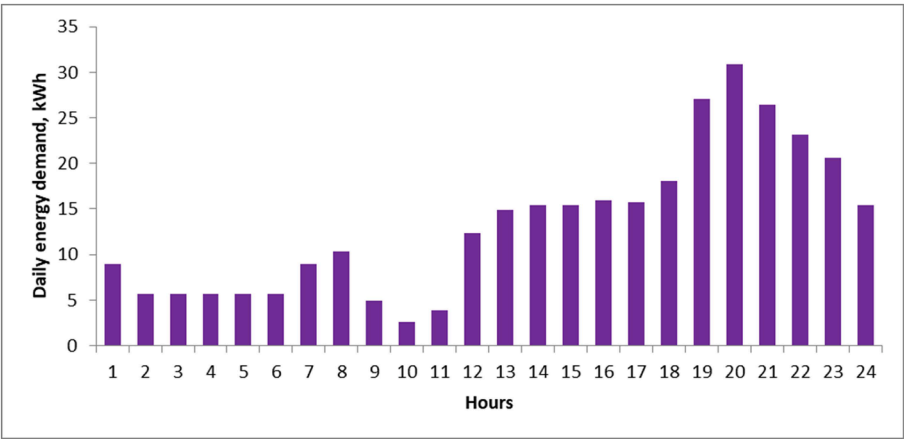


Fig. 4. Hourly energy demand profile.

The peak of electrical energy consumption was observed at 8 PM.

System architecture

Fig. 5 illustrates the architecture of the off-grid HRES under study. The HRES consists of a wind turbine, photovoltaic modules, batteries and converter. Wind turbines transform the kinetic energy of wind into electricity, in the form of alternating current (AC), while the photovoltaic panels convert solar radiation into direct current. The techno-economic parameters for each component in the architecture of the off-grid HRES will be evaluated and analysed in details in order to optimize the HRES proposed in this work.

System components model

PV array model

The photovoltaic (PV) cell used in this study was made from polycrystalline silicon. The light power is directly transformed into electricity directly by the solar cells. The PV modules are oriented to the south with an angle of inclination equal to 33° , which is very close to the optimum tilt angle in Morocco [1,15]). The power generated by the photovoltaic modules is computed using the following equation [11,30,33,36]:

$$P_{PV}(t) = Y_{PV} \times f_{rated} \times \left(\frac{G}{G_{ref}} \right) \times [1 + K_T(T_C - T_{ref})] \quad (1)$$

where Y_{PV} is the output power from PV module under standard test conditions (STC), G_{ref} and T_{ref} are the irradiance and temperature of PV module under STC, T_C is the temperature of PV module cell, f_{rated} is the derating factor of PV module and K_T is the power temperature coefficient.

The number of modules N_{PV_s} contained in a PV array was calculated

$$N_{PV} = N_{PV_s} \times N_{PV_p} \quad (2)$$

where N_{PV_s} is the number of PV module connected in series whereas N_{PV_p} is the number of PV module connected in parallel.

In our simulation, generic flat plat PV was chosen and the capital cost of 1 kW in the Moroccan market is considered to be equal to 1600 \$, the replacement price was taken equal to 0 \$ since the lifetime of the PV module is equal to the life of the project which is 25 years. The operating and maintenance price is taken equal to 10 \$/yr for each kW. The derating factor for the chosen PV module is equal to 80%.

Wind turbine model

For this study, a wind turbine with a horizontal axis was chosen. The wind turbine has nominal capacity 10 kW of alternating current (AC)

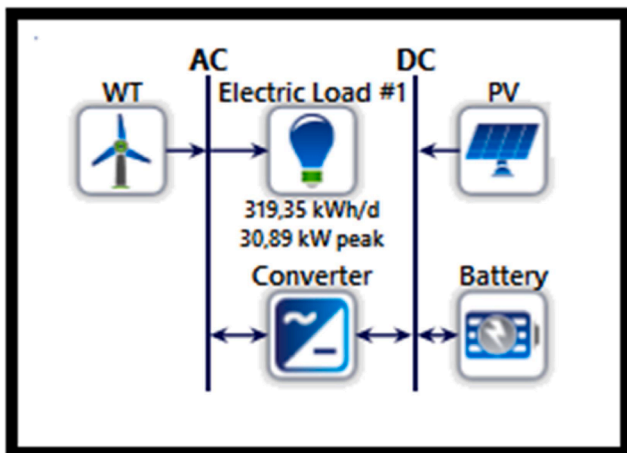


Fig. 5. Architecture of the autonomous HRES studied.

and more details about this wind turbine are shown in Table 2. The power generated $P_{WT}(t)$ by this wind turbine is calculated by the equation below [20,25]:

$$P_{WT}(t) = \frac{1}{2} \times \rho \times A_{ts} \times v(t)^3 \times C_{pc} \quad (3)$$

with

$$A_{ts} = \frac{\pi}{4} \times D_{WT}^2 \quad (4)$$

where ρ is the air density under standard condition (1.225 kg/m^3), A_{ts} is the swept area of the wind turbine rotor (m^2) and D_{WT} is the diameter of wind turbine rotor, v is the wind speed in (m/s) and C_{pc} is the power coefficient of the rotor.

The Bergey excel wind turbine was chosen in this work. It has a nominal power of 10 kW. Fig. 6 shows the wind turbine power curve. The cost of the wind turbine is 21,550 \$ and the cost of replacement and maintenance are 0 \$ and 100 \$/year, respectively.

Battery model

The HRES was selected to be autonomous (off-grid) in this work. The batteries are used to store electrical energy when there is an over production and a less demand. The energy stored in batteries must be used when the energy produced by both wind turbine and photovoltaic modules is insufficient to meet the energy demand of the village. In this work, lithium-ion batteries were selected due to their charge/discharge capacity and sustainable life cycle.

$$P_{batt} = \frac{D_{ele} \times N_d}{\eta_B \times \eta_{inv} \times D_d} \quad (5)$$

The storage capacity of the battery was estimated taking into account the number of days of autonomy (N_d), D_{ele} is the daily electrical charge (kWh/day), η_{inv} and η_B are the efficiency of the converter and the battery respectively and the depth of the battery discharge noted by D_d [32].

The battery selected in this study is Samsung M8068 P2 manufactured by Samsung, with a nominal capacity equal to 5.5 kW. The efficiency of the battery is 90%, the capital cost, replacement and maintenance are 1650 \$, 1650 \$, 15 \$/yr, respectively, as shown in Table 3.

Converter

The converter is used to convert direct current (DC) into alternating current (AC) or via versa. [16]. Table 4 shows the properties of the chosen converter.

System power balance model

The objective of this autonomous HRES is to generate electricity continuously at any time, t . The power generated by HRES and consumed by the load (demand) are determined as given by Eqs. (6) and (7).

$$P_{gen}(t) = P_{WT}(t) + P_{PV}(t) \quad (6)$$

$$P_{Dem} = P_{gen} \mp P_{batt} \quad (7)$$

Table 2
Specification data of WT.

Parameter	Specification
Model	Bergey windpower excel 10
Rated capacity (kW)	10
Rotor diameter (m)	7
Number of blades	3
Hub height (m)	30
Lifetime (yrs)	25
Capital cost (\$)	21,550
O & M cost (\$/yr)	100

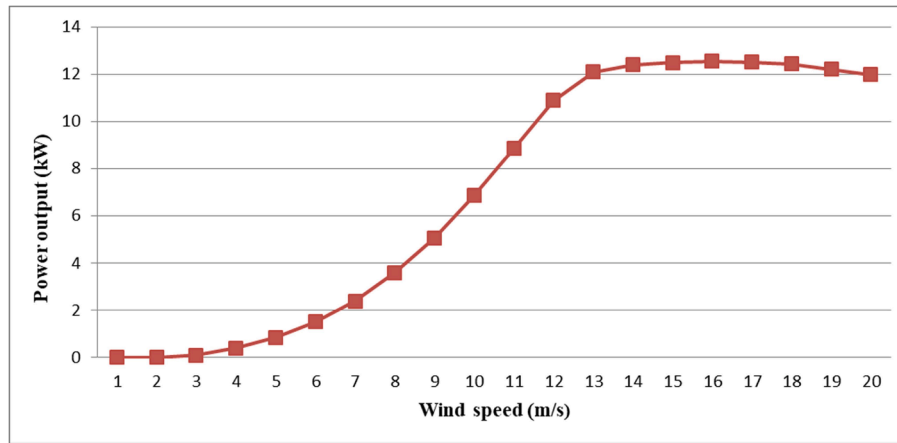


Fig. 6. WT power curve study.

Table 3
Specification of selected battery.

Parameter	Specification
Battery name	Samsung M8068 P2
Battery type	LI
Nominal voltage (V)	80.9
Nominal capacity (kWh)	5.5
Efficiency (%)	90
Throughput (kWh)	33,000
Capital cost (\$)	1650
Replacement cost (\$)	1650
O & M cost (\$/yr)	15

Table 4
Converter properties chosen in the study.

Parameter	Specification
Capital cost (\$/kW)	400
Replacement cost (\$/kW)	400
O&M costs (\$/kW/yr)	0
Efficiency (%)	95
Life time (yrs)	15
Relative capacity (%)	100

where P_{gen} , $P_{WT}(t)$, $P_{PV}(t)$, P_{batt} , P_{Dem} are generation power, generation power by wind turbine, generation power by solar PV, power storage in the battery and power consumed by the load, respectively.

It is convenient to indicate that the power generation undergoes the following conditions:

If $P_{gen}(t) > P_{Dem}(t)$, The surplus of electrical power generation is stored in the batteries.

If $P_{gen}(t) = P_{Dem}(t)$, $P_{batt} = 0$; The power generated by HRES is just enough to meet the load requested or energy demand.

If $P_{gen}(t) < P_{Dem}(t)$, and $P_{batt} > 0$; The power stored in the batteries compensates the shortage of power production by HRES to meet the load requested.

System simulation

The simulations were performed by Homer Pro software for the optimization, and techno-economic analyses of HRES. Homer determines the optimum configuration of the HRES taking into consideration the wind and solar resources, cost figures of each technology. Different configurations of HRESs were simulated and optimal system configuration, for each region was selected according to the lowest cost of energy (COE) and the net present cost (NPC) [24,37]. The database in the software contains several resources, load description, components

with their technical parameters, which are considered during each simulation.

The optimal case in Homer is found once the objective function (lowest NPC and COE) is satisfied. In this case, Homer Pro software evaluates the optimal case of the system by reducing the loss of power supply probability (LPSP) and the COE. To find the optimal configuration Homer starts calculating the number of wind turbines and the size of PV panels so that there is no lack of power supply to satisfy the electrical demand, if the electrical demand is not satisfied, Homer does not give any results and no configuration is proposed.

Once the meteorological, electrical and economic load data are entered, optimization constraints are entered in order to estimate the NPC and COE as if HOMER Pro were beginning to perform an economic analysis and rank the system configuration according to their NPC. Fig. 7 shows a simplified HOMER Pro comprehensive optimization framework.

Economic model and analysis

The optimum HRES system in this study was evaluated based on the net present cost (NPC) and cost of energy (COE) which are life-cycle economic indexes largely used in renewable energy project assessment.

The NPC and the COE are two very important criteria; the optimization is done according to the best configuration which corresponds to the minimum NPC. This NPC is a financial factor containing all the costs (e.g. installation cost, maintenance cost, replacement cost) minus the revenues realized by the hybrid system over the whole project life cycle. This makes the NPC and COE two factors of reference generation.

Net present cost

The NPC is an assessment made over the total project lifetime (25 years) of the costs incurred by the HRES minus the revenues reported by the system during its lifetime. NPC includes the capital, replacement, operation and maintenance cost, NPC is an important economic factor of HRES. The NPC can be determined as follows [8,12,14,34]:

$$NPC = \frac{C_{ann,tot}}{CRF \times (i, R_{proj})} \quad (8)$$

where $C_{ann,tot}$ is the total annual cost of the HRES (\$/yr), i is the actual discount rate (%), CRF is the capital recovery factor and R_{proj} is the project lifetime of the HRES (25 years).

Cost of energy (COE)

The COE of HRES is a ratio between annualized total cost (TAC) over total useful electricity produced by the system E_{pr} in (kWh/yr). The COE is mathematically expressed as follows [18]:

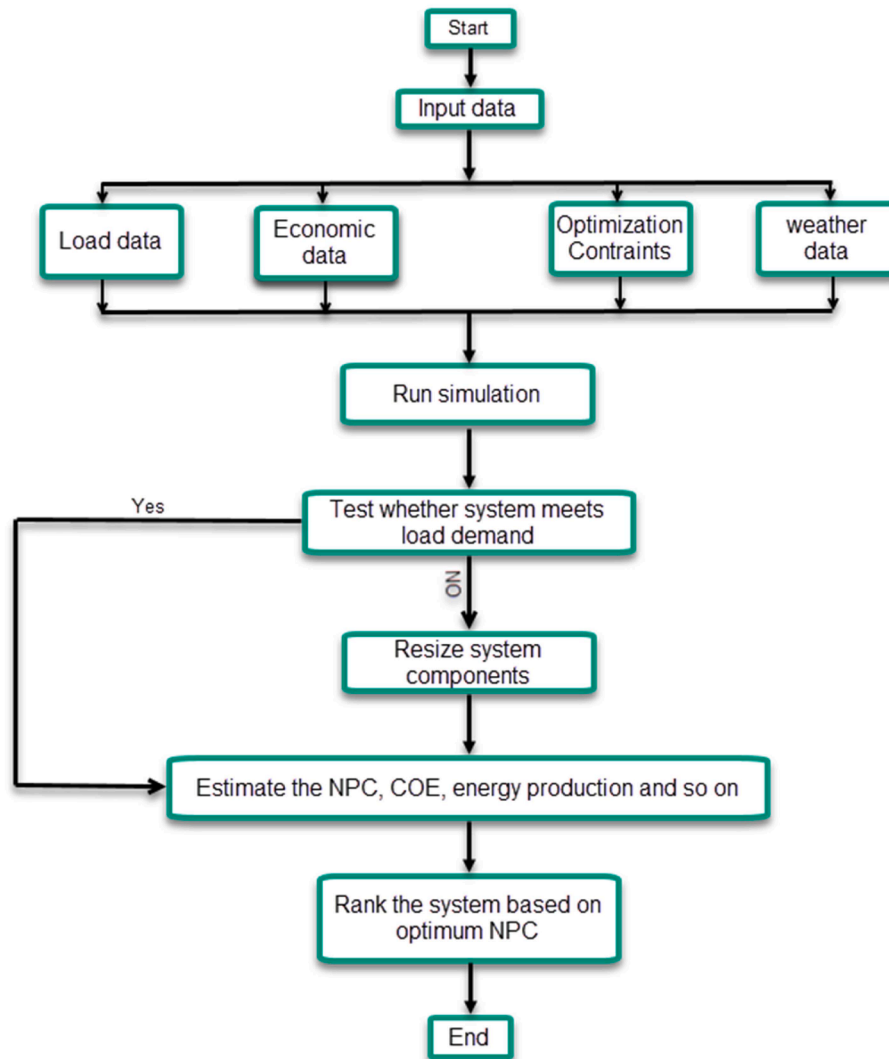


Fig. 7. A simplified global optimization framework for HOMER Pro.

$$COE = \frac{TAC}{E_{pr}} \quad (9)$$

Environmental impact

The potential of Wind-PV energy systems to reduce CO₂ emissions was estimated as follows keeping in mind that 1 kWh produced from HRES replaces 1 kWh generated by conventional energy system (conventional thermal power plant). Using Eq. (10), it is possible to calculate the quantities of CO₂ gas emissions eliminated by the HRES studied (in tones/year).

$$EM_{AV} = \frac{E_{PV}(F_{C_{grid}} - F_{C_{PV}}) + E_{WT}(F_{C_{grid}} - F_{C_{WT}})}{10^6} \quad (10)$$

Where $F_{C_{PV}}$ and $F_{C_{WT}}$ are emission factors related to PV and WT technologies, respectively. The HRES emission factor was estimated considering carbon footprint of renewable energy systems to be 50 g CO₂/kWh for PV, 13.7 g CO₂/kWh for wind technologies [2,31]. E_{PV} and E_{WT} represent the energy produced by the PV and WT power generation technologies, respectively. In Morocco the emission factor is considered equal to 746 g CO₂/kWh and is noted by $F_{C_{grid}}$ in the previous equation.

Results and discussion

In this work, an autonomous HRES has been designated to meet a same daily electrical energy demand (319.35 kWh) with a peak of 30.89 kW across 24 cities in Morocco. The load profile was considered to be constant in order to assess and rank the renewable energy potential of the selected cities. As stated earlier, the site selection respects both the climatic zoning of Morocco (6 zones) and the geographic location (12 regions).

The meteorological data for the 24 cities (Average solar radiation of 22 years) was obtained directly from the NASA Surface and Solar Energy website, using the exact geographic coordinates of the desired site. Fig. 8 shows the annual average solar irradiance (kWh/m²/day) and the clearness index for the all selected cities. The performance of PV modules is greatly influenced by solar irradiance [4,17,21,22]. The highest annual average of solar irradiance were recorded in the cities of Laayoune 5.97 kWh/m²/day, Tata 5.74 kWh/m²/day and Agadir 5.68 kWh/m²/day. Oppositely, the minimum recorded value of the annual average of solar irradiance was observed in the city of Oujda with 4.92 kWh/m²/day.

Also, the highest value of clearness index was obtained for Laayoune city with an annual average of 0.66 while the lowest value 0.58 was observed in Oujda.

Wind speed data was also obtained from the NASA surface and solar

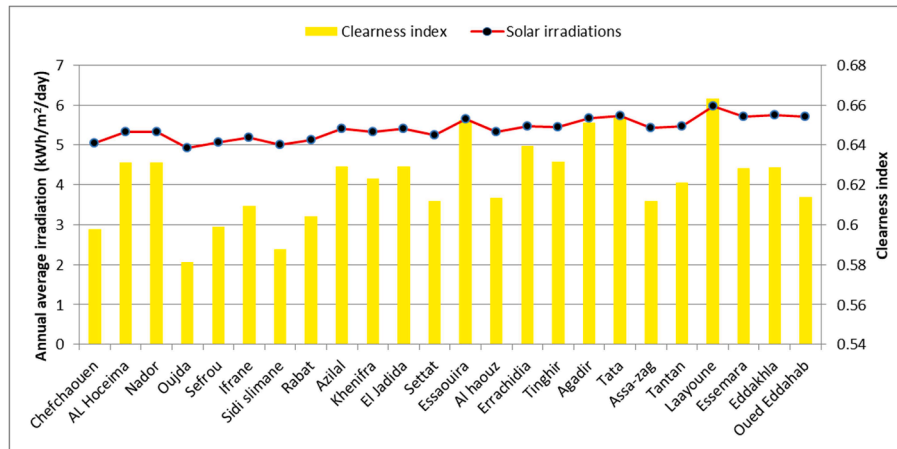


Fig. 8. Annual averages of solar irradiations and clearness index in selected cities.

energy website on the basis of a 10-year recording. Fig. 9 shows the annual average wind speed in all studied cities. A considerable variability can be noticed regarding wind speed profiles in these cities. It is illustrated that the annual average of the highest wind speeds were observed in the cities of Eddakhla (6.48 m/s), Laayoune (6.34 m/s) and Oued Eddahab (6.02 m/s). Whereas, the lowest annual average wind velocity was found in the city of Chefchaouen with a value of 3.15 m/s.

The HRESs for all 24 cities were optimized by the Homer pro software, taking into account as inputs, 25 years for the project lifetime period, 5% and 2.5% for the discount rate and the inflation rate respectively [5]. Solar PV data, wind turbine data, system components and their capital price, replacement cost and operating and maintenance cost are shown in the Table 5.

Homer pro performs many simulations, finds the optimal configurations and provides a complete energy balance for each configuration over the project period. The most viable solutions found by the software are ranked according to the lowest NPC and COE. A detailed financial presentation of capital, replacement, operation and maintenance costs are as well given in a life-cycle approach.

Optimal configuration

The main objective in this work is to study the feasibility of hybrid Wind-PV systems and examine the optimal configuration under the presence of two resources of renewable energy. Despite the fact that in some cities the optimal configuration may rely only on PV-Battery due to the poor potential of wind energy resources, the system architectures

Table 5
Information on the components of HRES.

	PV (1 kW)	WT (10 kW)	Converter (1 kW)	Battery (5.5 kWh)
Capital cost (\$)	1600.00	21,550.00	400.00	1650.00
Replacement cost (\$)	0.00	0.00	400.00	1650.00
O&M (\$/yr)	10.00	100.00	0.00	15.00
Efficiency (%)	—	—	95	90

consisting of two technologies (PV and wind) was considered in the first simulation phase for ranking purposes. Later, the most suitable scenarios based on only PV-battery technologies in specific sites are discussed.

From the simulation results generated, Fig. 10 shows the optimal configuration of the Wind -PV-battery for each city. The largest capacity of photovoltaic array recorded in 24 cities is 256 kW noticed in the city of Oujda with only 1 wind turbine, 34.3 kW for the converter and 107 batteries. In second place comes the town of Chefchaouen with a capacity of 167 kW of PV array, 1 wind turbine, 50.9 kW inverter and 157 batteries. On the other hand, the city of Nador has the lowest PV panel capacity with a PV capacity of 64.7 kW, 8 wind turbines, a 41.1 kW inverter and 81 batteries. In other towns, the capacity of the PV array varies between 142 kW for Sidi slimane and 68.1 kW for Laayoune.

It is clear from Fig. 10 that Nador is the city with the highest number of wind turbines with a total of 8 units, followed by Essaouira with 6 turbines, Settat with 5 turbines and then the other towns have less than 4

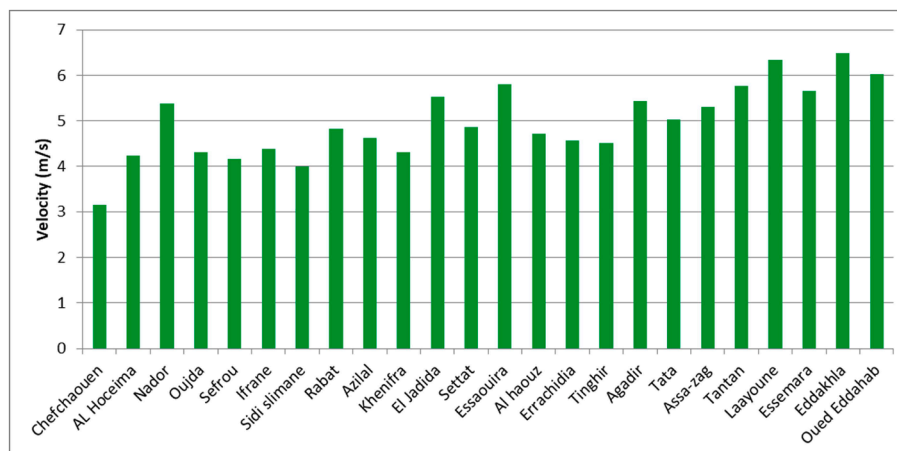


Fig. 9. Annual average of wind speed in selected cities.

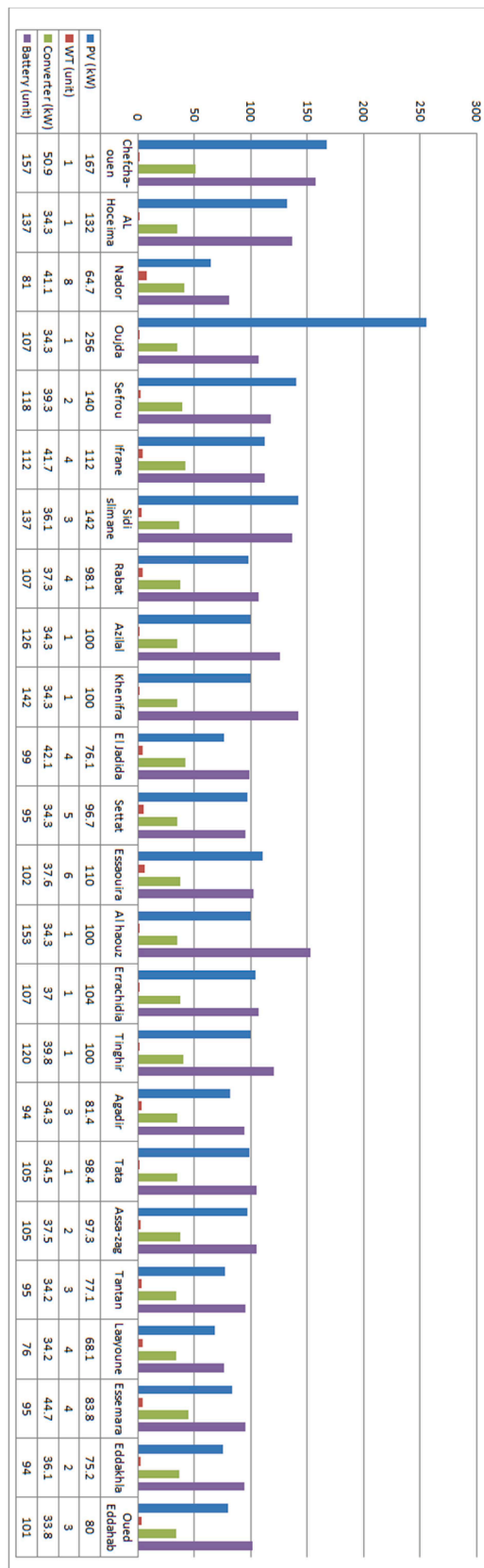


Fig. 10. The optimal PV-Wind configuration of all cities.

wind turbines.

Despite the participation of wind turbines in electricity production to meet the energy demand recorded in some cities, their overall contribution remains low to certain extent. This can be attributed to the expensive price of wind turbines along with the low potential of wind energy resource.

Figs. 11a and 11b show respectively, the cost of energy, NPC and capital cost of the hybrid system for each city, corresponding to the optimal configurations illustrated in Fig. 9. In this graph, the most suitable location to host such systems from an economic point of view is Eddakhla with a COE, NPC and capital cost equal to 0.171 \$/kWh, 370,078 \$ and 332,973 \$, respectively. The same results were approximately obtained in Laayoune city with a COE, NPC and capital cost of 0.172 \$/kWh, 370,984 \$ and 334,195 \$, respectively. The city with the most expensive electricity cost is appeared is Oujda with a value of 0.32 \$/kWh. The corresponding NPC and capital cost are found to 691,478 \$ and 621,408 \$, respectively.

The cost of electricity generated by the Wind-PV-Battery system in other cities varies between 0.184 \$/kWh and 0.299 \$/kWh, these extreme values were observed in the cities of Tantan and Chefchaouen, respectively.

Electricity production for optimal PV-Wind configuration of all cities

Fig. 12a displays the average monthly generation of the optimal Wind-PV system for all the studied cities. A general observation is that the production of electricity generated by photovoltaic array dominates in the majority of the studied cities. There are cities where wind energy production does not exceed 5% such as Chefchaouen, Oujda, Azilah, khenifra, Al Haouz, Errachidia, Tinghir, Tata and Sefrou as shown in Fig. 12b.

The cities of Sidi Slimane, Ifrane, rabat, Assa-zag and Settat can be considered as moderate in terms of electricity production from wind turbines. In the remaining cities, the participation of wind in the production of electricity is relatively significant especially in the cities of Laayoune and Nador with the percentage of production reaching 40.70% and 50.60%, respectively.

Based on the results presented earlier, it was concluded that some cities present marginal portion of wind energy contribution and accordingly the COE was found significantly high. As such, it can be stated that PV-battery systems are more desirable in these sites because the COE drops significantly as shown in Table 6.

Table 7 shows the comparison between the optimum PV-Wind-Battery and PV-Battery configuration for the 9 cities. From this table it is clear that the energy price and the NPC for the optimal PV-battery configuration is lower than the COE and NPC of the configuration which contains both energy sources at the same time in all cities.

The increase in the COE is 6.78% and 2.08% in the cities of Chefchaouen and Sefrou respectively. It is for this reason that in the 9 cities the installation of the wind system is not profitable.

Mitigation potential of CO₂ emissions

Fig. 13 displays the amount of CO₂ that can be saved by adopting WT-PV HES in the different regions in Morocco. These saving CO₂ emission results can help to assess environmental and climatic advantages of these WT-PV HES installations in a specific location. The amount of mitigated CO₂ emissions differs depending on the renewable energy technology and its contribution to the overall energy production as discussed in previous section. Oujda represents the maximum quantity of CO₂ saving of 291.4 tonnes of CO₂ for WT-PV technologies. Oppositely, the cities that produce the minimum CO₂ emissions mitigation potential are Khenifra and Al Haouz with 126.71 tonnes of CO₂ and 126.59 tonnes of CO₂, respectively. The HRES in all studied 24 cities have the potential to save a large amount of CO₂ estimated to be 3,711.42 tonnes of CO₂.

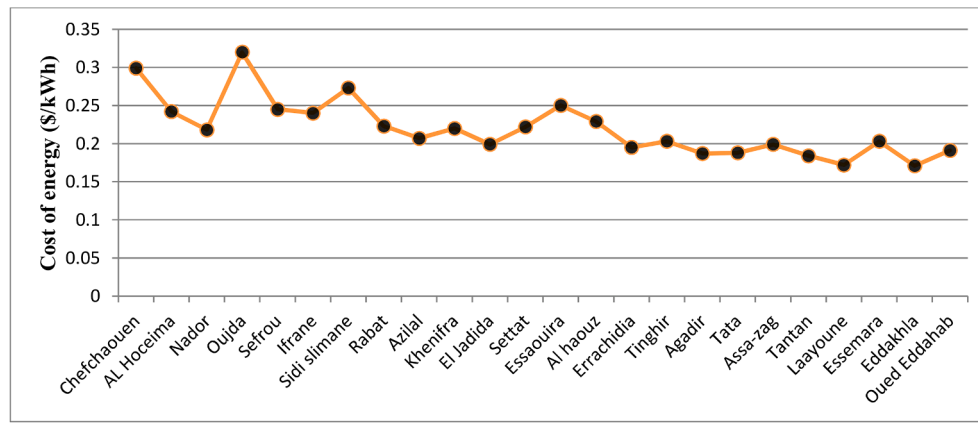


Fig. 11a. COE for the optimal PV-Wind system for the 24 cities.

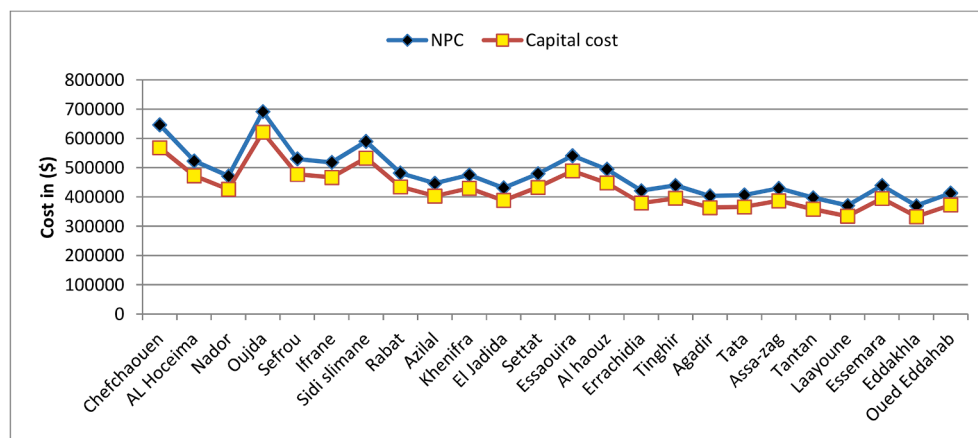


Fig. 11b. NPC and capital cost for the optimum PV-Wind system for the 24 cities.

Cost reduction effect

As mentioned in previous sections, the current market of HRES causes that the COE is not yet competitive against electric grid cost. In this section, a sensitivity analysis is practiced to inspect how the outputs are affected by the price reduction rate in the capital cost of the HRES. The reduction rate in the capital cost has decreased gradually from 0% to 30% with a step of 10%.

The effect of the price reduction on the price of electricity generated by the HRES is illustrated in Fig. 14. It is obvious that the cost of electricity tends to decrease in all 24 cities. To exemplify, it is reduced from 0.171 to 0.125 \$/kWh for Eddakhla and Laayoune. In the city of Oujda where the highest energy cost was appeared, a capital cost reduction of 30% causes a drop of the COE from 0.32 to 0.235 \$/kWh.

Social dimension

Installing HRES based on wind, solar PV and battery for 24 cities in different climate region in Morocco has many impacts on the social, economic and the life habitats of people in these cities. The development of any community basically depends on the social and economic aspects. Many social advantages can be gained when using wind and solar PV technologies in different climate regions for 24 cities in Morocco. These social advantages have benefits for the communities living in these regions such as creating job opportunities, enhancing the health level, and improving standard of living and level of education. HRES can promote as well the agriculture especially for rural areas (Wind Solar Hybrid Renewable Energy System, 2020). The study carried out by [27] shows

that the activity of the citizens in the region where the renewable production system installed grow significantly. The impact of rural electrification on the income of citizen and the energy expenditure also shown in the study done in Nepal [40]. The citizen income increases by 8% while the energy expenditure saved by 50% when the families connect to the electricity compared to families not connected to electricity. The service of electricity connection by HRES in isolated and remote area helps the woman in general to participate in the socio-economic life in these cities, the schooling of the girls which allows increasing the enrolments in the schools and also reduces the illiteracy rate in addition to child improvement and maternal health.

Conclusion

The energy, economic and environmental (3E) study for sustainable electrification was proposed on the national scale in Morocco. The feasibility of off-grid HRES for wind-PV system was evaluated extensively for a total 24 cities. These cities are distributed in twelve regions based on the geographical location of the country with two cities for each region.

The NASA metrological data for wind speed and solar radiation was used. The simulations were performed by HOMER Pro software. Both key indexes performance for financial and environmental of HRES have been evaluated for different twelve regions of Morocco. Eddakhla and Laayoune cities have been demonstrated to the best site according to the lowest COE 0.171 and 0.172 \$/kWh, respectively.

The percentage of electrical energy production from wind turbine and solar PV technologies for these cities was varied depending on the



Fig. 12a. Electrical power generation of the optimum PV-Wind configuration of all cities.

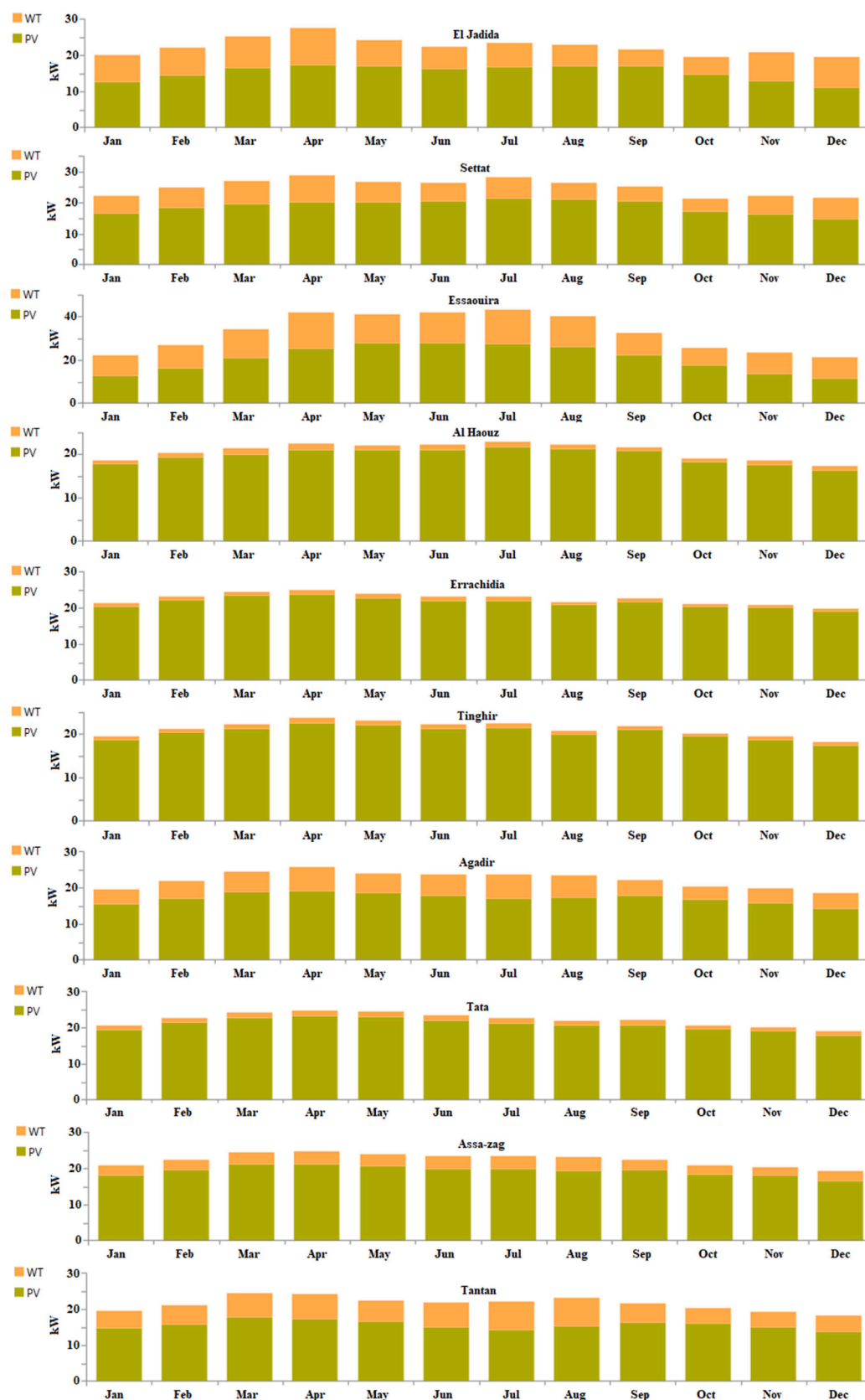


Fig. 12a. (continued).

prevailing weather data. Laayoune city has 40% of power production from WT and 59.3% power production from PV, while the Eddakhla city has 25.6% of power production from WT and 74.40% power production

from PV. The amount of CO₂ per year that can be saved in Laayoune and Eddakhla cities are about 154.21 and 128.21 tonnes, respectively. The Nador city shows the percentage of power production by WT is higher

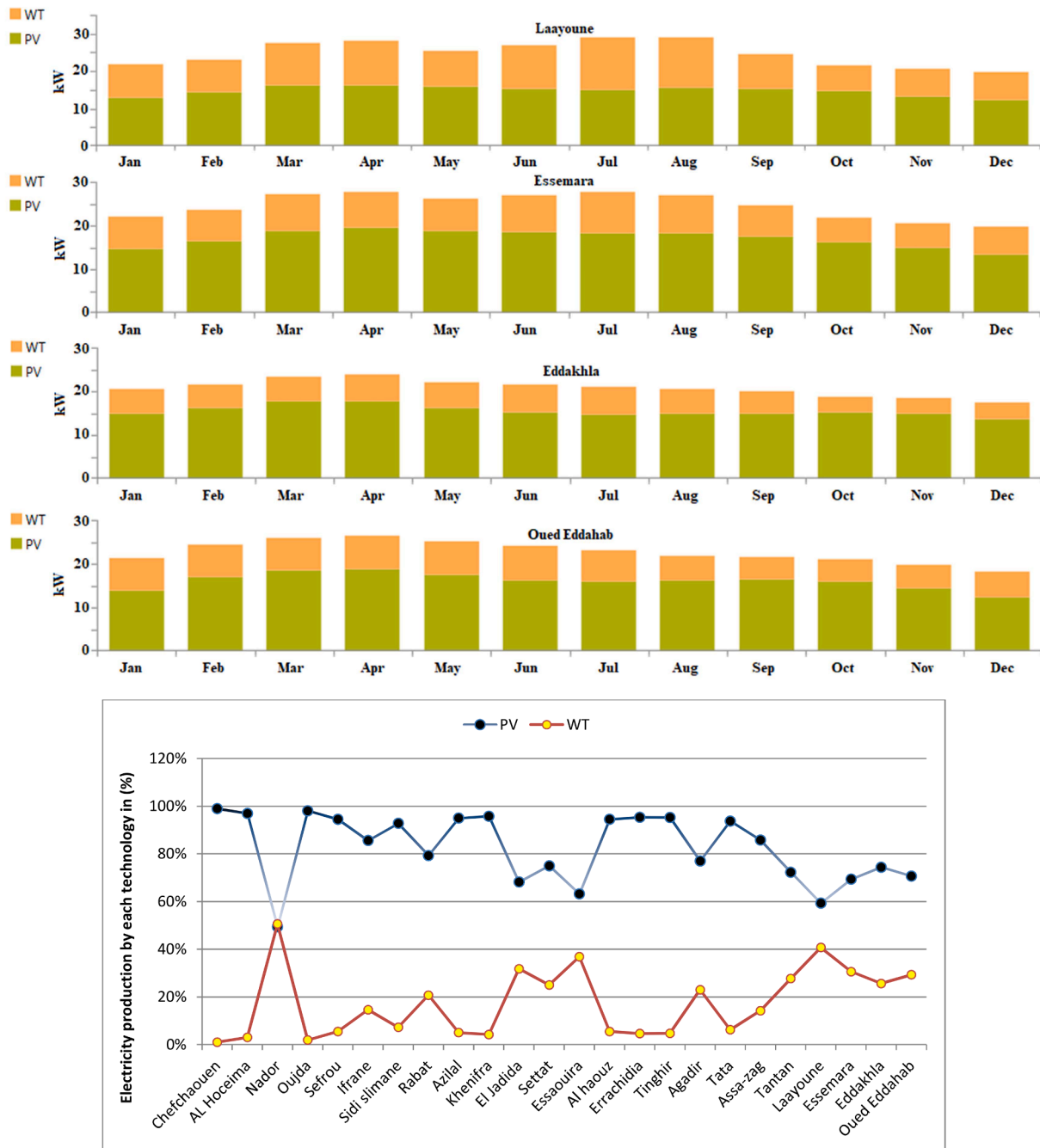


Fig. 12b. The production of electricity by each technology in the selected cities.

Table 6

Information on the optimal PV-Battery system in 9 cities.

Cities	Optimal configuration	COE (\$/kWh)	NPC (\$)	Capital cost (\$)
Chefchaouen	PV-Battery	0.28	620,043	545,399
Oujda	PV-Battery	0.285	616,489	545,820
Sefrou	PV-Battery	0.24	519,095	466,181
Ifrane	PV-Battery	0.239	516,338	463,390
Azilal	PV-Battery	0.199	431,146	388,460
Khenifra	PV-Battery	0.217	468,887	423,625
Al Haouz	PV-Battery	0.222	480,233	430,523
Errachidia	PV-Battery	0.192	414,380	372,543
Tinghir	PV-Battery	0.199	430,115	387,007

Table 7

Comparison of the optimum PV-Battery and PV-Wind-Battery configuration in the 9 cities.

PV-Battery	PV-Wind-Battery			
Cities	COE (\$/kWh)	NPC (\$)	NPC (\$)	COE (\$/kWh)
Chefchaouen	0.28	620,043	646,843	0.299
Oujda	0.285	616,489	691,478	0.32
Sefrou	0.24	519,095	530,068	0.245
Ifrane	0.239	516,338	517,897	0.24
Azilal	0.199	431,146	446,712	0.207
Khenifra	0.217	468,887	475,156	0.22
Al Haouz	0.222	480,233	494,711	0.229
Errachidia	0.192	414,380	421,658	0.195
Tinghir	0.199	430,115	439,366	0.203

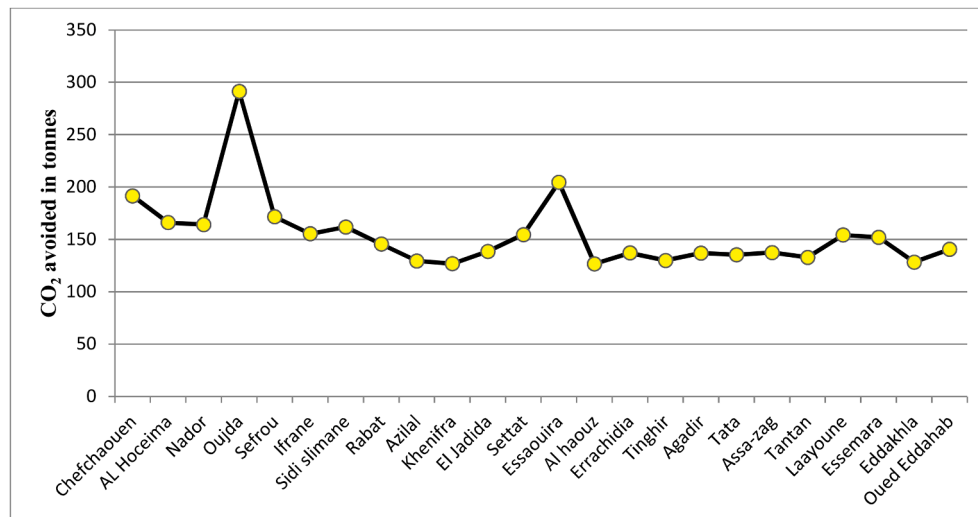
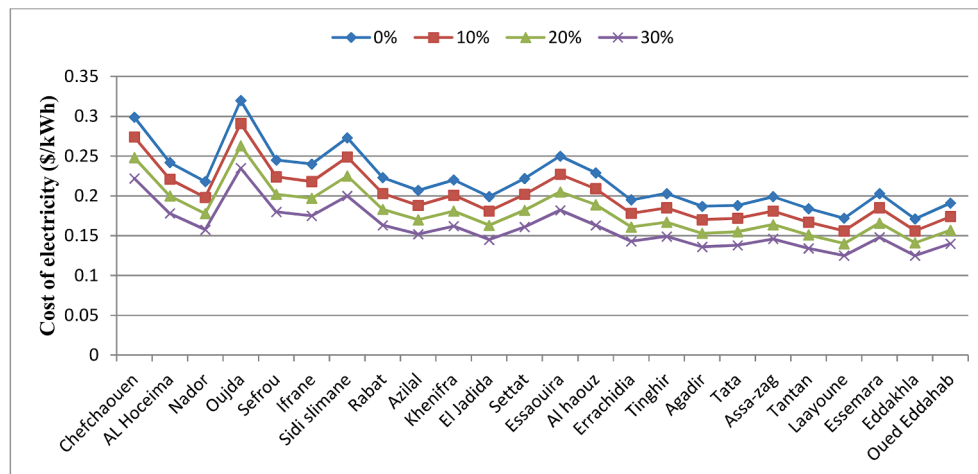
Fig. 13. CO₂ emissions avoided in the cities sampled.

Fig. 14. Effect of cost reduction on the COE.

than percentage of power production by PV, whereas, some sites are not suitable for wind energy production.

This genre of projects is eco-friendly and will allow Morocco to increase its energy independence by relying on renewable energy resources. Furthermore, generalizing the use of such systems in isolated towns and villages which are not connected to the electrical networks in the African continent can engender substantial social and economic benefits.

Author Contributions

Conceptualization, methodology and software: H. EL-HOUARI and A. ALLOUHI. Revision: T. Salameh. Visualization and supervision A. JAMIL, B. EL AMRANI and T. KOUSKSOU.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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